



Dissertation on
Impact of Generative AI Usage Among Working
Professionals in the Service Industry

Interim Report

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Declaration

This document has been prepared by the student of PES University, Bengaluru, as an Interim Report for the Executive Master of Business Administration (MBA) – Capstone Project under the specialization in Data Science and Business Analytics.

This report is submitted in partial fulfilment of the academic requirements of Semester V of the Executive MBA program. A significant portion of this interim report is expected to be included in the final project submission, with suitable revisions and updates based on further analysis and findings.

The project titled **Impact of Generative AI Usage Among Working Professionals in the Service Industry** has been undertaken as part of the prescribed academic curriculum and represents the original work carried out by the undersigned during the study.

I hereby declare that the content of this report is original, has not been submitted to any other institution for the award of any degree or diploma, and complies with the academic standards and regulations of the Department of Management Studies, PES University.

Acknowledgement

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Abstract

The adoption of Generative Artificial Intelligence (AI) tools is rapidly increasing among working professionals in the service industry, supporting tasks such as information gathering, writing, documentation, technical assistance, and decision-making. While these tools offer potential benefits in improving productivity and efficiency, their real impact on work outcomes, employee satisfaction, and perceived risks is not yet fully understood.

This study aims to analyse how service industry professionals use Generative AI tools and to evaluate their perceived impact on time savings, task efficiency, accuracy improvement, workload reduction, and problem-solving ability. The study also examines key challenges associated with AI usage, including data privacy concerns, uncertainty regarding AI-generated outputs, and skill-related difficulties, and assesses how these factors influence overall user satisfaction.

Primary data for this research is being collected through an online survey administered to working professionals in the service industry. The collected data will be analysed using descriptive statistics, regression analysis, and clustering techniques to identify drivers of satisfaction and to segment users based on AI adoption behaviour. As an interim report, this document presents the research design and proposed analytical framework, with final findings to be reported upon completion of data analysis.

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1. Introduction

Industry Profile

The service industry is one of the most knowledge-intensive and people-driven sectors, encompassing domains such as information technology, healthcare, finance, consulting, customer support, education, and professional services. In this sector, employee productivity, quality of work, and timely decision-making play a critical role in organizational performance. With increasing workload complexity, tighter deadlines, and growing client expectations, service organizations are actively exploring digital technologies to enhance efficiency and reduce manual effort.

In recent years, Generative Artificial Intelligence (AI) has emerged as a transformative technology within the service industry. Tools powered by Generative AI are increasingly used for tasks such as information retrieval, content creation, documentation, technical assistance, problem-solving, and decision support. These tools have the potential to significantly improve work efficiency, accuracy, and speed while reducing repetitive and time-consuming tasks.

The adoption of Generative AI in the service industry is influenced by multiple external factors, which can be understood using a PESTLE framework:

- **Political:** Government policies and national AI strategies influence enterprise-level AI adoption, particularly in regulated service sectors.
- **Economic:** Organizations are under pressure to improve productivity and reduce operational costs, making AI-driven efficiency gains economically attractive.
- **Social:** There is a growing acceptance of AI-assisted work among professionals, along with expectations for faster and smarter work outputs.
- **Technological:** Rapid advancements in Generative AI models, cloud computing, and automation tools have accelerated AI adoption across service roles.
- **Legal:** Data protection regulations and compliance requirements govern how AI tools can access, process, and store organizational and customer data.
- **Environmental:** Digital solutions, including AI-driven workflows, support reduced paper usage and optimized resource consumption, contributing to sustainability goals.

The primary objective of this project is to analyse how working professionals in the service industry use Generative AI tools and to assess their impact on productivity, efficiency, accuracy, workload reduction, and overall satisfaction. The study also seeks to identify risks and challenges associated with AI adoption, enabling organizations to adopt these technologies responsibly and effectively.

Peer Analysis

Peer Application 1: ChatGPT (OpenAI)

Model Used: Large Language Models (LLMs) for Generative AI assistance

STP Model:

- Segmentation: Knowledge workers, professionals, students, and technical users across service domains.
- Targeting: Individuals seeking quick information, writing assistance, problem-solving support, and learning guidance.
- Positioning: “AI-powered assistant for productivity, creativity, and problem-solving.”

ChatGPT is widely used by service industry professionals for tasks such as drafting emails and reports, generating ideas, understanding technical concepts, and supporting decision-making. Its strength lies in its conversational interface, versatility across use cases, and ability to reduce cognitive and time burdens for users.

Peer Application 2: Microsoft Copilot

Model Used: Generative AI integrated with enterprise productivity tools

STP Model:

- Segmentation: Corporate and enterprise users within structured organizational environments.
- Targeting: Professionals using Microsoft tools such as Word, Excel, Outlook, and Teams.
- Positioning: “AI-powered productivity assistant embedded within everyday work tools.”

Microsoft Copilot focuses on enhancing productivity within enterprise workflows by integrating Generative AI directly into commonly used applications. It supports tasks such as document drafting, data summarization, meeting insights, and workflow automation, with a strong emphasis on enterprise security and compliance.

This study draws insights from both standalone Generative AI tools and enterprise-integrated AI platforms. By analysing real-world usage patterns and employee perceptions, the project aims to identify key drivers of efficiency and satisfaction while addressing challenges related to trust, skills, and responsible AI adoption in the service industry.

2. Literature Review

This section reviews selected academic studies and industry reports related to the use of Generative Artificial Intelligence (AI) in the service industry. The literature highlights key objectives, methodologies, and findings related to productivity improvement, efficiency gains, and challenges associated with AI adoption in professional environments.

1. **Brynjolfsson, E., Rock, D., & Syverson, C. (2023). *Generative AI at Work*.**

This study examines the impact of Generative AI tools on employee productivity using a field experiment approach. The findings indicate that AI assistance improves task completion speed and work quality, particularly for less-experienced employees.

Source: National Bureau of Economic Research (NBER).

2. **Dwivedi, Y. K., et al. (2023). *Opportunities and challenges of Generative AI*.**

The objective of this study is to analyse the benefits and risks of Generative AI adoption in organizations. The study highlights efficiency improvement as a key benefit, while data privacy and reliability of AI outputs are identified as major concerns.

Source: International Journal of Information Management.

3. **McKinsey Global Institute. (2023). *The economic potential of Generative AI*.**

This report evaluates the productivity potential of Generative AI across service industries using task-based analysis. The findings suggest that AI can reduce time spent on repetitive tasks and enhance overall work efficiency.

Source: McKinsey & Company.

4. **Microsoft Work Trend Index. (2023). *Will AI Fix Work?***

This industry report analyses professional usage of Generative AI tools through large-scale surveys. The results show improved productivity and workload reduction, along with concerns related to trust and data security.

Source: Microsoft.

From the studies reviewed, the following observations can be made:

- Generative AI tools positively influence productivity and efficiency in service roles.
- Employee satisfaction is affected by both perceived benefits and risk concerns.
- Data privacy, reliability, and skill gaps remain key challenges in AI adoption.
- There is a need for survey-based empirical studies to understand real-world AI usage, which this study aims to address.

3. Research Methodology

Research Problem

The rapid adoption of Generative Artificial Intelligence (AI) tools in the service industry has the potential to significantly influence employee productivity, work efficiency, and overall satisfaction. However, the actual impact of AI usage on daily work outcomes, along with associated risks such as data privacy concerns, reliability issues, and skill limitations, is not clearly understood.

This study addresses the problem of understanding how Generative AI tools are used by working professionals and how these tools affect efficiency, workload, accuracy, and user satisfaction in service-oriented roles.

Data Collection and Preparation

Primary data for this study is being collected through a structured online survey designed using Google Forms. The survey targets working professionals across various service industry domains. It captures information related to respondent profiles, AI usage frequency, tasks supported by AI, perceived productivity improvements, risk factors, and overall satisfaction.

The survey uses a combination of categorical variables, multiple-choice responses, and 0–10 scale ratings to measure perceived impact and challenges. Data collected from the survey will be exported into a structured dataset for further analysis.

Data Preparation

- The dataset consists of survey responses collected from working professionals in the service industry. Incomplete or invalid responses were removed to ensure data consistency and reliability.
- Categorical variables such as work experience, AI usage frequency, and overall satisfaction were standardized to maintain uniform response categories.
- Multi-select responses related to AI-supported tasks were converted into dummy variables to enable quantitative analysis.
- Rating-based responses measured on a 0–10 scale were validated and, where required, transformed into scores to ensure consistency across metrics.
- Missing values were handled using appropriate techniques, such as mode imputation for categorical variables and median imputation for scaled responses.
- Data types were verified and corrected to ensure suitability for statistical analysis and machine learning models.

Data Summary

Dataset Size: 113 records and 15 features

Data Source: Primary data collected through an online Google Forms questionnaire

Data Types: Categorical, ordinal (0–10 scale), binary, and numerical variables

Missing Values: Minimal and addressed during data preparation

Key Variables:

- Work Experience (Categorical)
- AI Usage Frequency (Categorical)
- Tasks supported by AI (Multi-select categorical)
- Time saved by using AI (Categorical)
- Efficiency improvement (0–10 scale → % score)
- Accuracy improvement (0–10 scale → % score)
- Workload reduction (0–10 scale → % score)
- Problem-solving improvement (0–10 scale → % score)
- Privacy concern, reliability uncertainty, skill difficulty (Risk factors)
- Overall satisfaction (Yes/No)

Secondary Data

Secondary data for this study was collected from industry reports, research articles, case studies, and publicly available documentation related to the use of artificial intelligence in the service sector. These sources were used to support the research background, validate findings, and provide contextual understanding of AI adoption trends.

Descriptive Statistics

Categorical variables

Experience_Level Distribution:		
	Count	Percentage (%)
2-5 years	34	30.09
5-10 years	32	28.32
10-15 years	23	20.35
0-2 years	15	13.27
15-20 years	7	6.19
20+ years	2	1.77

The respondent base is strongly concentrated in the 2–10 years’ experience range (58.41%), indicating that AI adoption is highest among professionals who are actively balancing productivity demands and skill development. Lower adoption among highly experienced professionals (20+ years: 1.77%) may suggest resistance to change or established work routines, while early-career professionals (0–2 years: 13.27%) may still be exploring foundational skills rather than relying heavily on AI tools.

Usage_Frequency Distribution:		
	Count	Percentage (%)
Daily	70	61.95
Weekly	31	27.43
Monthly	6	5.31
Rarely	5	4.42
Never	1	0.88

With nearly 62% daily users, AI tools appear to function as a core operational aid rather than an occasional support tool. The extremely low proportion of non-users (0.88%) indicates near-universal awareness and acceptance, while the weekly usage segment (27.43%) suggests a secondary group that leverages AI selectively for specific tasks rather than continuous workflows.

Daily_Time_Saved Distribution:

	Count	Percentage (%)
30-60 minutes	39	34.51
Less than 30 minutes	37	32.74
1-2 hours	19	16.81
2-3 hours	8	7.08
More than 4 hours	4	3.54
No time saved	3	2.65
3-4 hours	3	2.65

Time savings are not marginal but structurally meaningful, as over 26% of respondents save more than one hour per day, which can translate into significant cumulative productivity gains at an organizational level. The small percentage reporting no time savings (2.65%) suggests that inefficiency concerns are minimal, and AI tools generally deliver positive efficiency outcomes across diverse user groups.

Overall_Satisfaction Distribution:

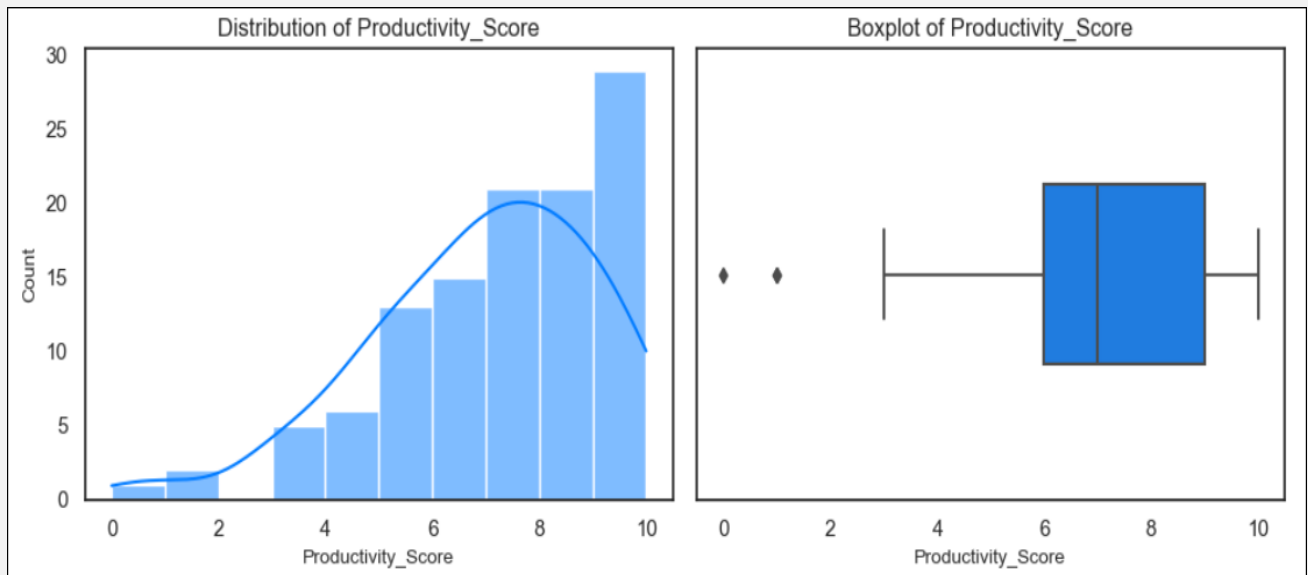
	Count	Percentage (%)
Yes	96	84.96
No	17	15.04

The high satisfaction rate (84.96%) reinforces that frequent usage and tangible time savings directly translate into positive user perception. The presence of a dissatisfied segment (15.04%) indicates that while AI tools are effective overall, usability, accuracy, or trust-related factors may still require improvement to achieve universal satisfaction.

Numerical Variables

	count	mean	std	min	25%	50%	75%	max
Productivity_Score	113.0	6.929204	2.141081	0.0	6.0	7.0	9.0	10.0
Workload_Reduction_Score	113.0	6.628319	2.155536	1.0	5.0	7.0	8.0	10.0
Problem_Solving_Score	113.0	6.584071	2.222955	0.0	5.0	7.0	8.0	10.0
Accuracy_Score	113.0	6.778761	2.218971	0.0	5.0	7.0	8.0	10.0
Data_Privacy_Concern_Score	113.0	7.176991	2.749124	0.0	6.0	8.0	10.0	10.0
Uncertainty_Score	113.0	6.194690	1.865372	0.0	5.0	6.0	8.0	10.0
Skill_Difficulty_Score	113.0	4.504425	2.921973	0.0	2.0	4.0	7.0	10.0
UseCase_Information_Research	113.0	0.433628	0.497783	0.0	0.0	0.0	1.0	1.0
UseCase_Writing_Support	113.0	0.353982	0.480334	0.0	0.0	0.0	1.0	1.0
UseCase_Document_Creation	113.0	0.203540	0.404424	0.0	0.0	0.0	0.0	1.0
UseCase_Media_Assistance	113.0	0.115044	0.320497	0.0	0.0	0.0	0.0	1.0
UseCase_Technical_Assistance	113.0	0.247788	0.433651	0.0	0.0	0.0	0.0	1.0
UseCase_Problem_Solving	113.0	0.221239	0.416930	0.0	0.0	0.0	0.0	1.0
UseCase_Other	113.0	0.000000	0.000000	0.0	0.0	0.0	0.0	0.0

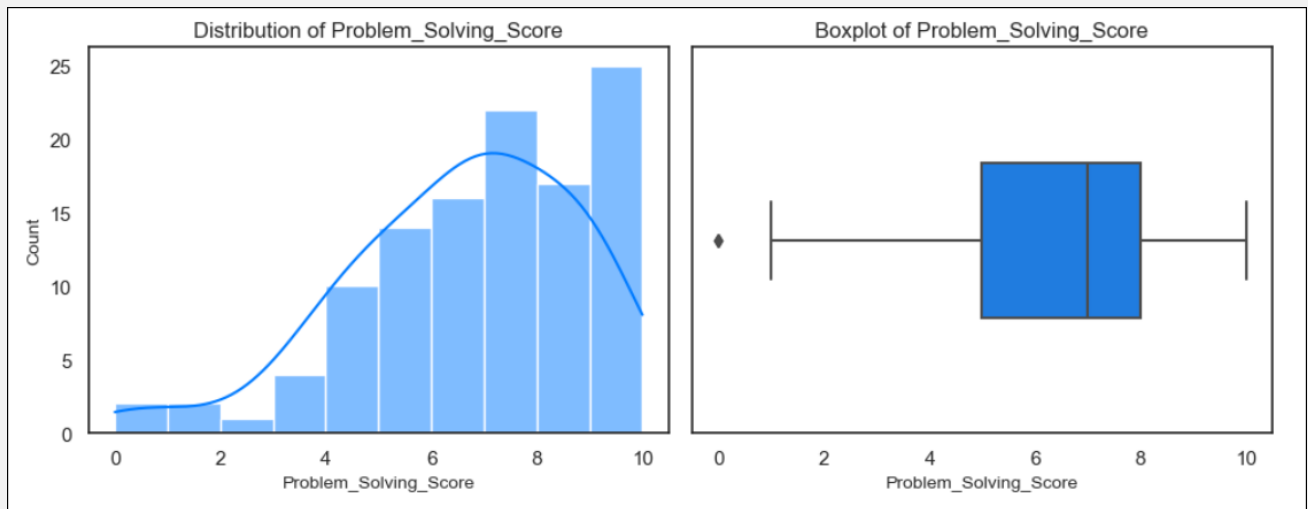
Mean scores for productivity, workload reduction, problem solving, and accuracy are consistently above 6.5, indicating that AI tools generally have a positive impact on work efficiency and task performance. Median values ranging between 7 and 8 across key performance metrics reflect strong agreement among respondents regarding the benefits of AI usage. At the same time, data privacy concern scores remain relatively high, suggesting that trust and data security continue to be important considerations despite widespread adoption. The comparatively lower skill difficulty scores indicate that users do not perceive AI tools as difficult to use, supporting their accessibility across different experience levels. Finally, usage patterns show that AI tools are primarily leveraged for information research and writing support, while other use cases exhibit comparatively lower adoption.



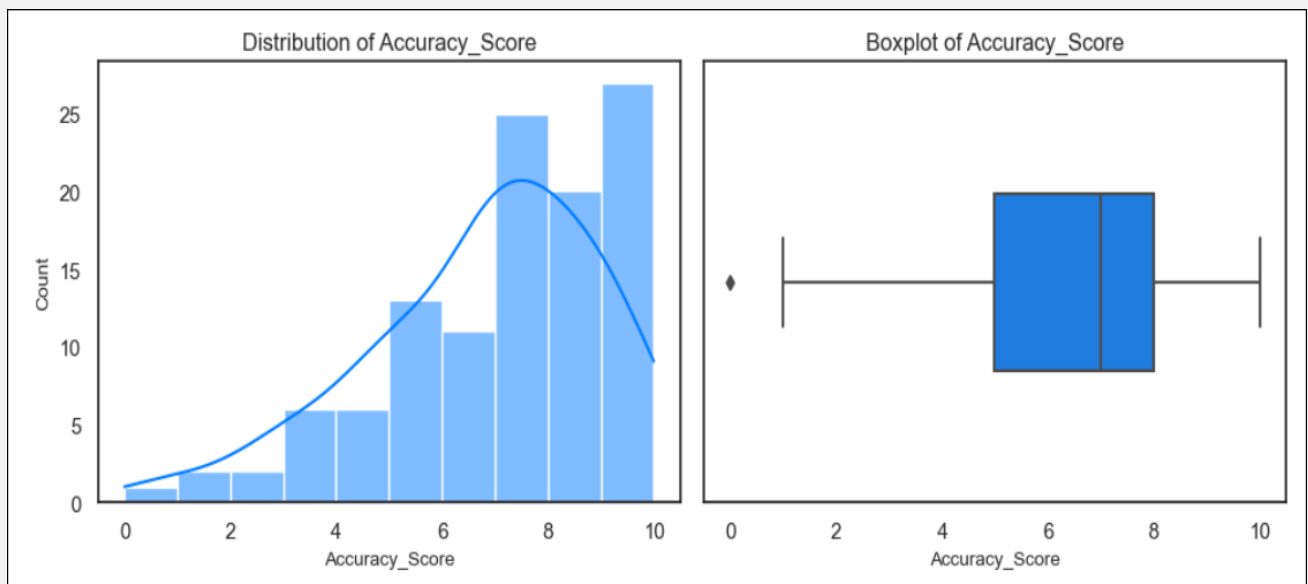
The distribution of the Productivity_Score is moderately left-skewed, with a higher concentration of responses between 6 and 9, indicating that most users perceive AI tools as having a strong positive impact on productivity. The boxplot shows a compact interquartile range with only a few low-end outliers, suggesting generally consistent productivity gains across respondents with limited extreme negative experiences.



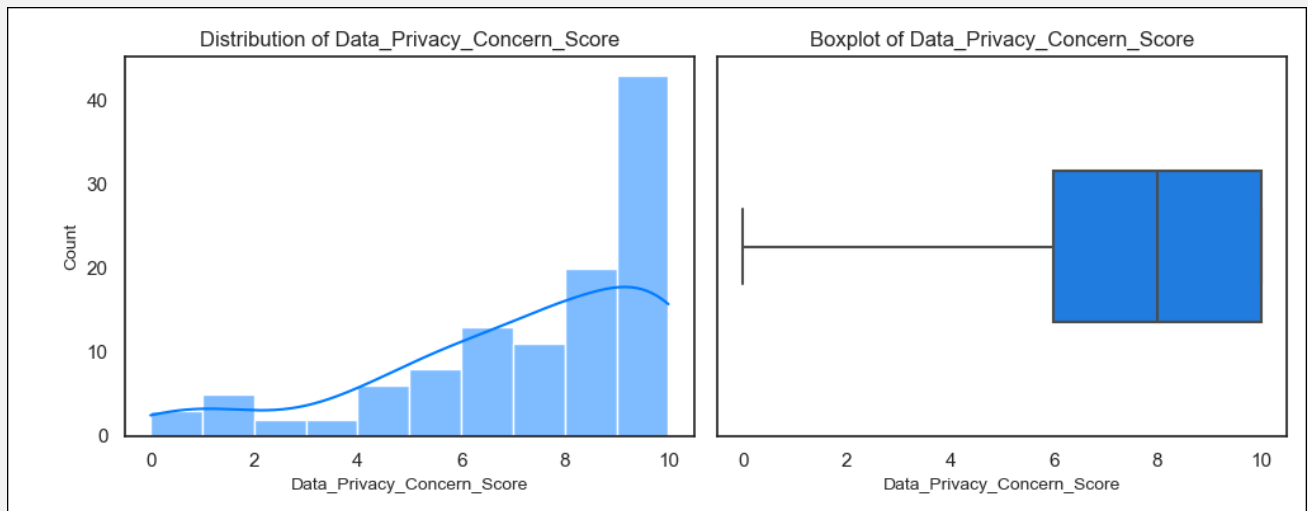
The Workload_Reduction_Score is concentrated between 6 and 8, indicating that most users perceive AI tools as effective in reducing workload. The boxplot shows limited variability and few outliers, suggesting consistent workload reduction experiences across respondent



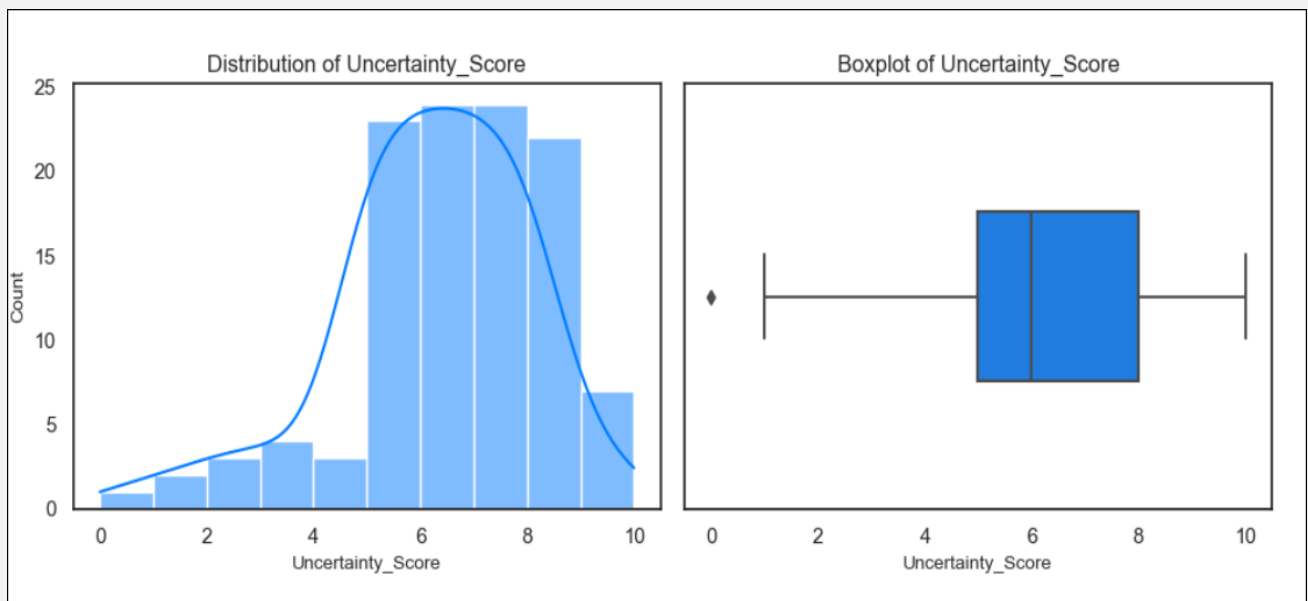
The Problem_Solving_Score distribution is centered around higher values (approximately 6–9), indicating that most users find AI tools effective for problem-solving tasks. The boxplot shows moderate variability with a few low-end outliers, suggesting generally positive perceptions with limited negative experiences.



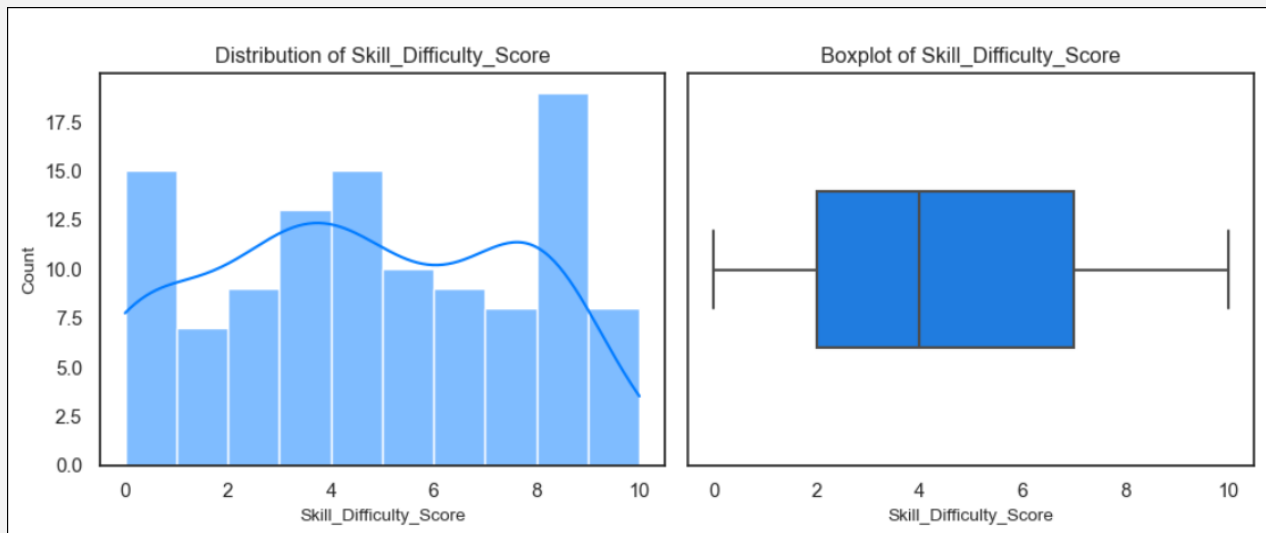
The Accuracy_Score distribution is skewed toward higher values, with most responses concentrated between 6 and 9, indicating that users generally perceive AI outputs as accurate. The boxplot shows a relatively narrow interquartile range with only a few low-end outliers, suggesting consistent accuracy perceptions across respondents.



The Data_Privacy_Concern_Score is heavily concentrated toward higher values (7–10), indicating that most users have strong concerns regarding data privacy when using AI tools. The boxplot shows a wide spread with a few low-end outliers, suggesting that while concerns are generally high, a small group of users is less worried about privacy.



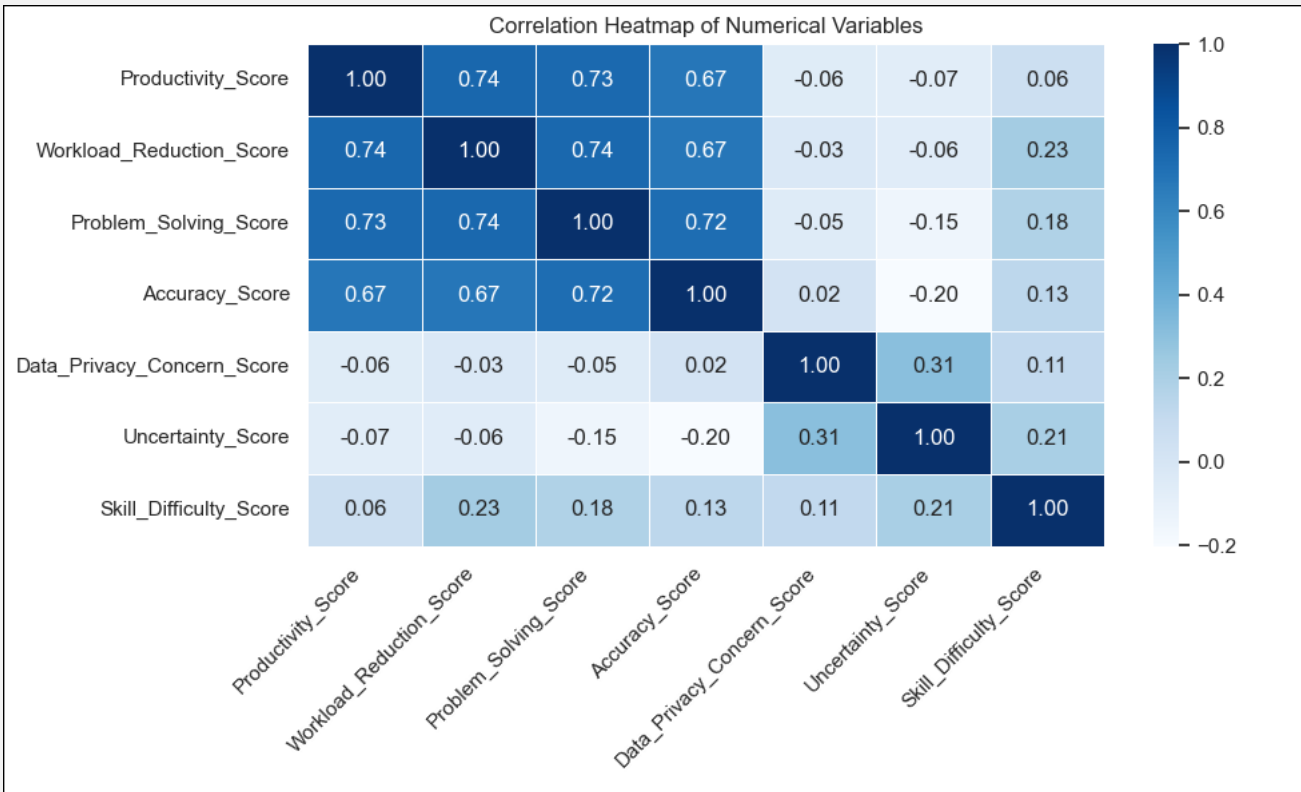
The Uncertainty_Score is mainly concentrated between 5 and 8, indicating a moderate level of uncertainty among users regarding AI outcomes. The boxplot shows moderate variability with a small number of low-end outliers, suggesting that while most users feel reasonably confident, a few experiences higher uncertainty.



The Skill_Difficulty_Score distribution is spread across the scale but centred around lower to mid values (3–6), indicating that most users do not find AI tools highly difficult to use. The boxplot shows moderate variability with no extreme outliers, suggesting relatively consistent perceptions of skill difficulty among respondents.

Bivariant Analysis

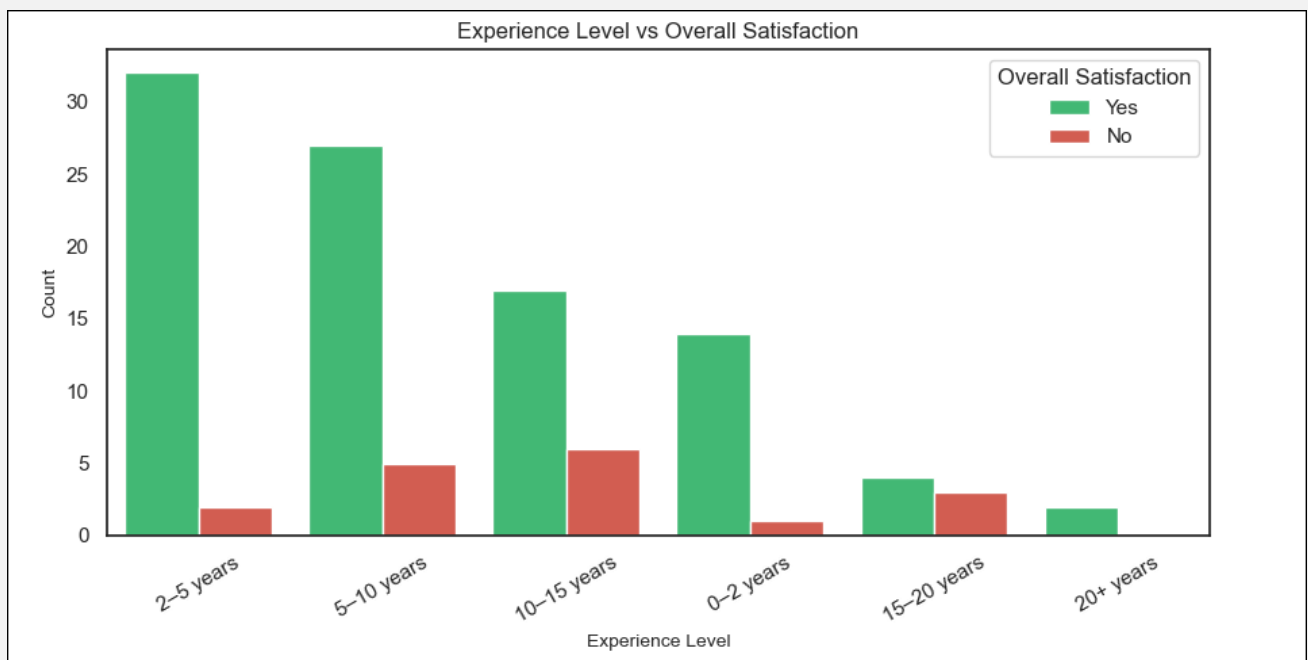
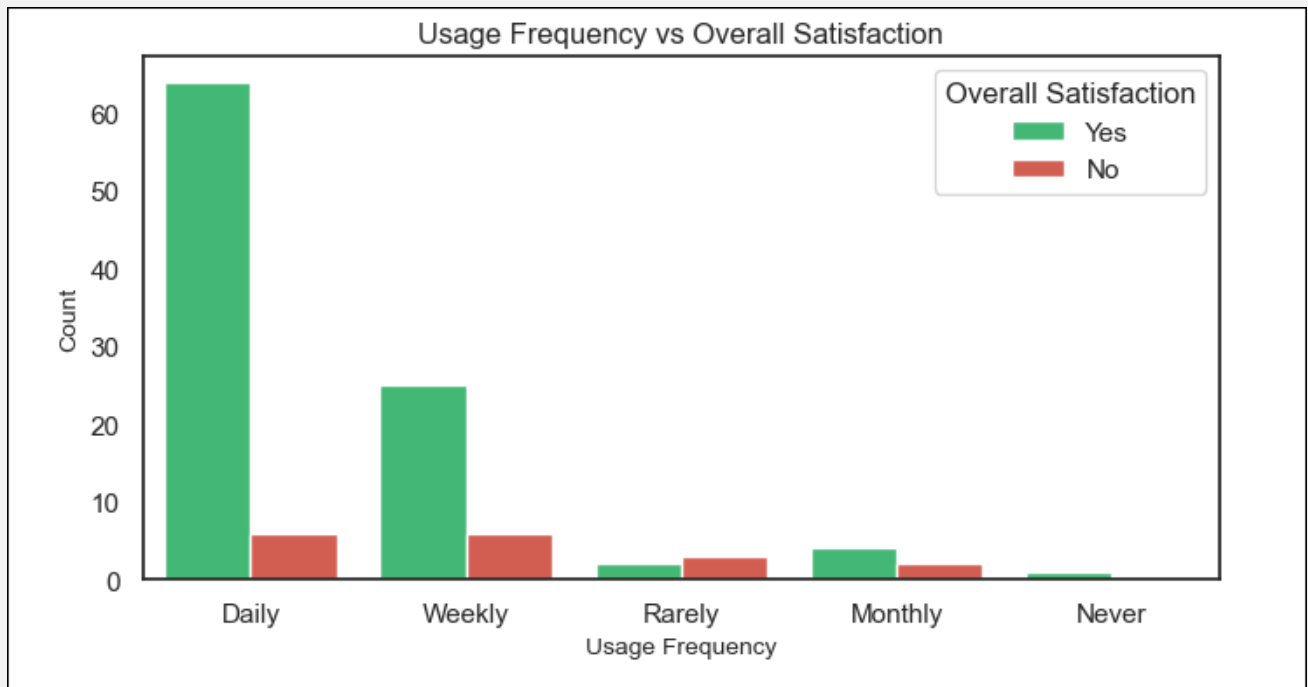
Numerical Variables



The correlation analysis shows a strong positive relationship among Productivity, Workload Reduction, Problem Solving, and Accuracy scores (correlation coefficients ranging from approximately 0.67 to 0.74), indicating that improvements in one performance dimension are closely associated with gains in others. In contrast, Data Privacy Concern and Uncertainty scores exhibit weak or negative correlations with productivity-related metrics, suggesting that user concerns and uncertainty are largely independent of perceived performance benefits. Skill difficulty shows only weak correlations with all variables, implying that ease of use does not strongly influence perceived productivity or accuracy outcomes.

Categorical variables

Regular users show significantly higher satisfaction, with daily and weekly users overwhelmingly reporting “Yes” compared to “No,” indicating strong perceived value from frequent AI usage. In contrast, rare and infrequent users exhibit relatively higher dissatisfaction, suggesting that limited engagement may reduce perceived benefits.



Satisfaction levels are highest among mid-career professionals (2–10 years), where positive responses (“Yes”) significantly outweigh dissatisfaction, indicating strong perceived value of AI tools in this group. In contrast, satisfaction declines slightly at higher experience levels, suggesting that very senior professionals may perceive comparatively fewer benefits from AI adoption.

Model Development

Train–Test Split

The dataset was divided into training and testing sets using a 70:30 split to evaluate model performance on unseen data. The training set consisted of 79 records with 17 features, and the test set consisted of 34 records with 17 features. This split ensured sufficient data for learning while maintaining a test sample for evaluation.

Model Comparison

Multiple classification models were trained and evaluated to identify the most suitable approach for predicting Overall Satisfaction. All models were trained on the same training data and evaluated using Accuracy and Weighted F1-score to ensure fair comparison.

```
Logistic Regression:  
Accuracy = 0.8529, F1 = 0.8260  
  
Decision Tree:  
Accuracy = 0.7647, F1 = 0.7523  
  
Random Forest:  
Accuracy = 0.8529, F1 = 0.8260  
  
KNN:  
Accuracy = 0.7647, F1 = 0.7339  
  
Naive Bayes:  
Accuracy = 0.5882, F1 = 0.6085
```

Based on overall performance and consistency, Logistic Regression and Random Forest emerged as the strongest models.

Final Model Selection

Although Random Forest matched the accuracy of Logistic Regression, Logistic Regression was selected as the final model due to:

- Comparable performance with simpler model structure
- Better interpretability of results
- Suitability for small and structured datasets

This makes Logistic Regression more practical for explanation and business interpretation.

Logistic Regression – Model Results

=== Logistic Regression Results ===				
Accuracy: 0.8529411764705882				
Classification Report:				
	precision	recall	f1-score	support
0	1.00	0.38	0.55	8
1	0.84	1.00	0.91	26
accuracy			0.85	34
macro avg	0.92	0.69	0.73	34
weighted avg	0.88	0.85	0.83	34
Confusion Matrix:				
[[3 5]				
[0 26]]				

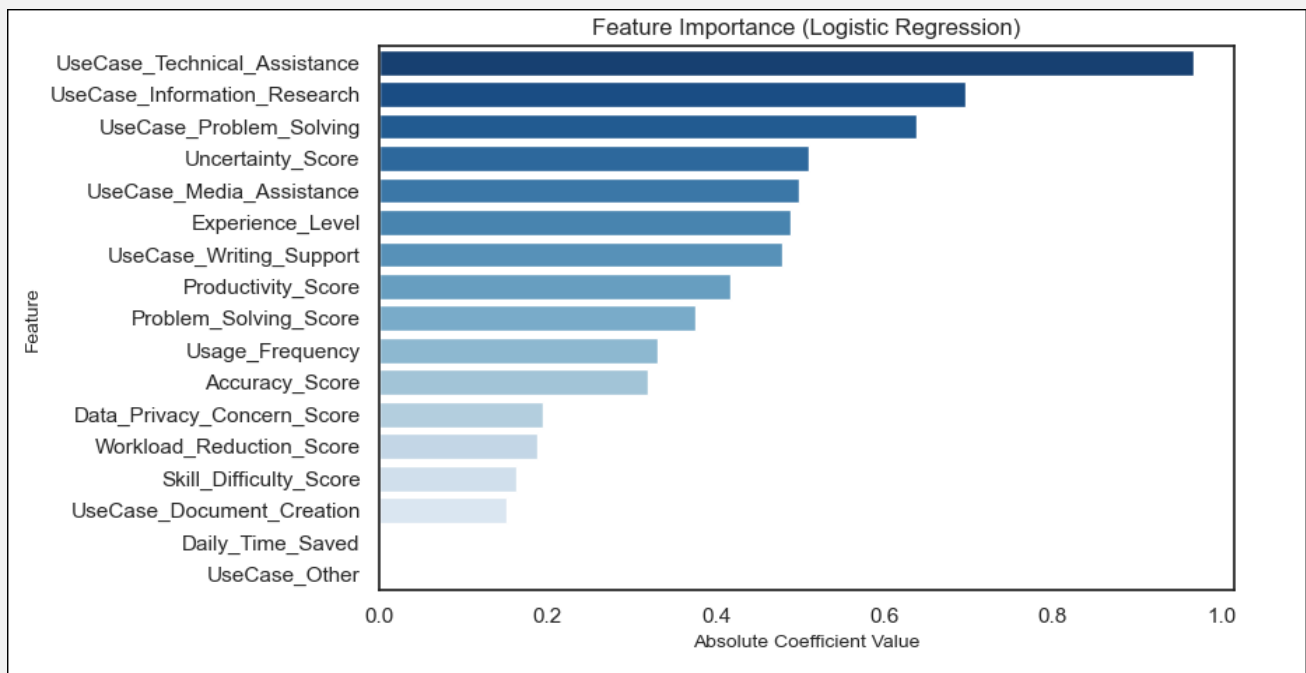
The initial Logistic Regression model achieved an accuracy of 85.29%. The model demonstrated a high recall for the positive class (class 1), indicating a strong ability to correctly identify satisfied cases. However, the recall for the negative class (class 0) was comparatively lower, suggesting scope for improvement in identifying unsatisfied cases. The confusion matrix further confirmed this behaviour, showing that while most positive cases were correctly classified, a portion of negative cases were misclassified.

Model Improvement Using Threshold Optimization

Best threshold for class 0 recall: 0.7				
Accuracy: 0.9117647058823529				
Classification Report:				
	precision	recall	f1-score	support
0	0.86	0.75	0.80	8
1	0.93	0.96	0.94	26
accuracy			0.91	34
macro avg	0.89	0.86	0.87	34
weighted avg	0.91	0.91	0.91	34
Confusion Matrix:				
[[6 2]				
[1 25]]				

To improve model performance, probability-based prediction was applied using Logistic Regression instead of relying on the default classification threshold of 0.5. Multiple threshold values were evaluated to enhance class-level performance, and an optimal threshold of 0.7 was identified. This adjustment resulted in a better balance between the classes and improved overall prediction quality. After threshold optimization, the model achieved an accuracy of 91.18%. The recall for class 0 increased significantly from 0.38 to 0.75, while the model continued to perform strongly for class 1 with a recall of 96%. The updated confusion matrix reflected a noticeable reduction in misclassifications across both classes.

Feature Importance



The feature importance analysis indicates that UseCase_Technical_Assistance is the most influential factor in predicting overall satisfaction, followed by UseCase_Information_Research and UseCase_Problem_Solving, highlighting the importance of practical and task-oriented usage. Features such as Uncertainty_Score, Experience_Level, and Productivity_Score show moderate influence, suggesting that user confidence and efficiency also affect satisfaction. In contrast, factors like Data_Privacy_Concern_Score, Skill_Difficulty_Score, and UseCase_Document_Creation have relatively lower impact, indicating a weaker direct relationship with satisfaction in this dataset.

The model was developed in Python, leveraging:

- scikit-learn for modelling and validation
- pandas and NumPy for data preprocessing
- matplotlib and seaborn for visualization and exploratory analysis

Final Outcomes

Threshold-optimized Logistic Regression provided the best overall performance, achieving higher accuracy and better class balance without increasing model complexity. This final model was selected for deployment and interpretation.

The expected outcome of this project is to develop a reliable predictive model that can accurately estimate overall user satisfaction based on usage patterns and experience-related factors. The model is expected to help identify key drivers of satisfaction, support data-driven decision-making, and provide insights that can be used to improve system effectiveness and user experience.

Conclusion

In this project, multiple classification models were developed and evaluated to predict overall satisfaction. After comparing model performance, Logistic Regression was selected due to its strong accuracy, interpretability, and consistent results. Further improvement was achieved through threshold optimization, which enhanced class balance and increased overall accuracy. Feature importance analysis revealed that task-oriented use cases and technical assistance play a significant role in influencing satisfaction. The final model provides meaningful insights and demonstrates the effectiveness of data-driven approaches in understanding user behaviour.

Recommendations

Based on the model findings, greater focus should be placed on enhancing technical assistance and problem-solving capabilities, as these features have the highest impact on user satisfaction. Improving support for information research and practical usage scenarios can further increase positive outcomes. Continuous monitoring of user feedback and periodic model updates are recommended to maintain prediction accuracy over time.

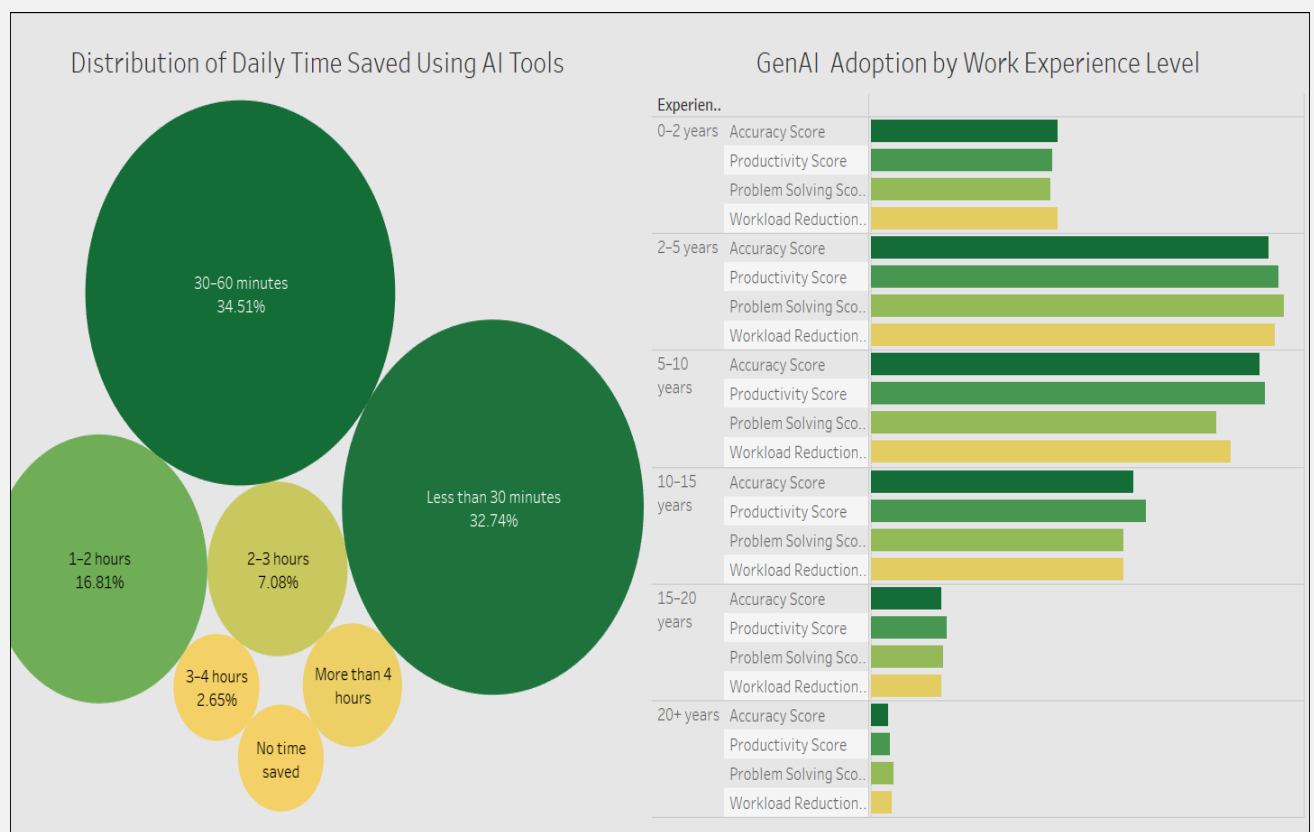
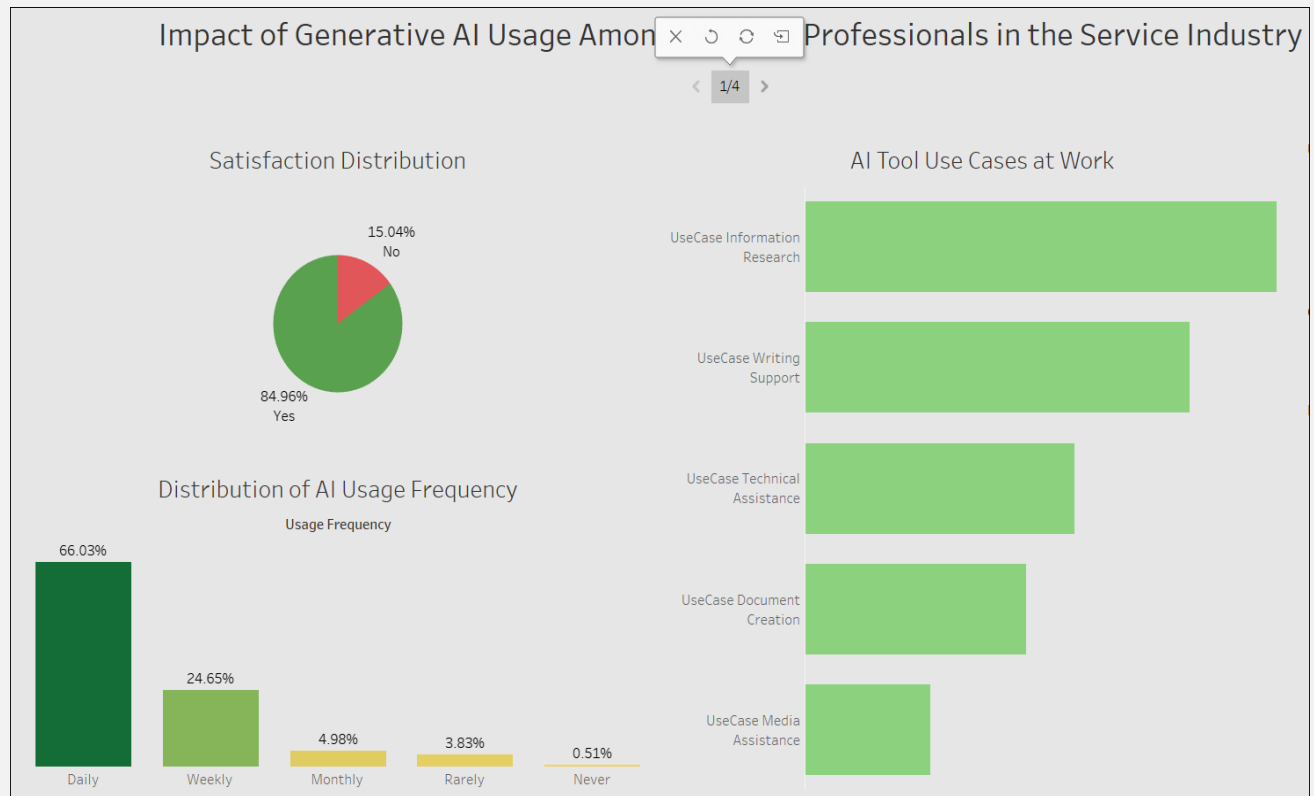
Scope of Future Enhancements

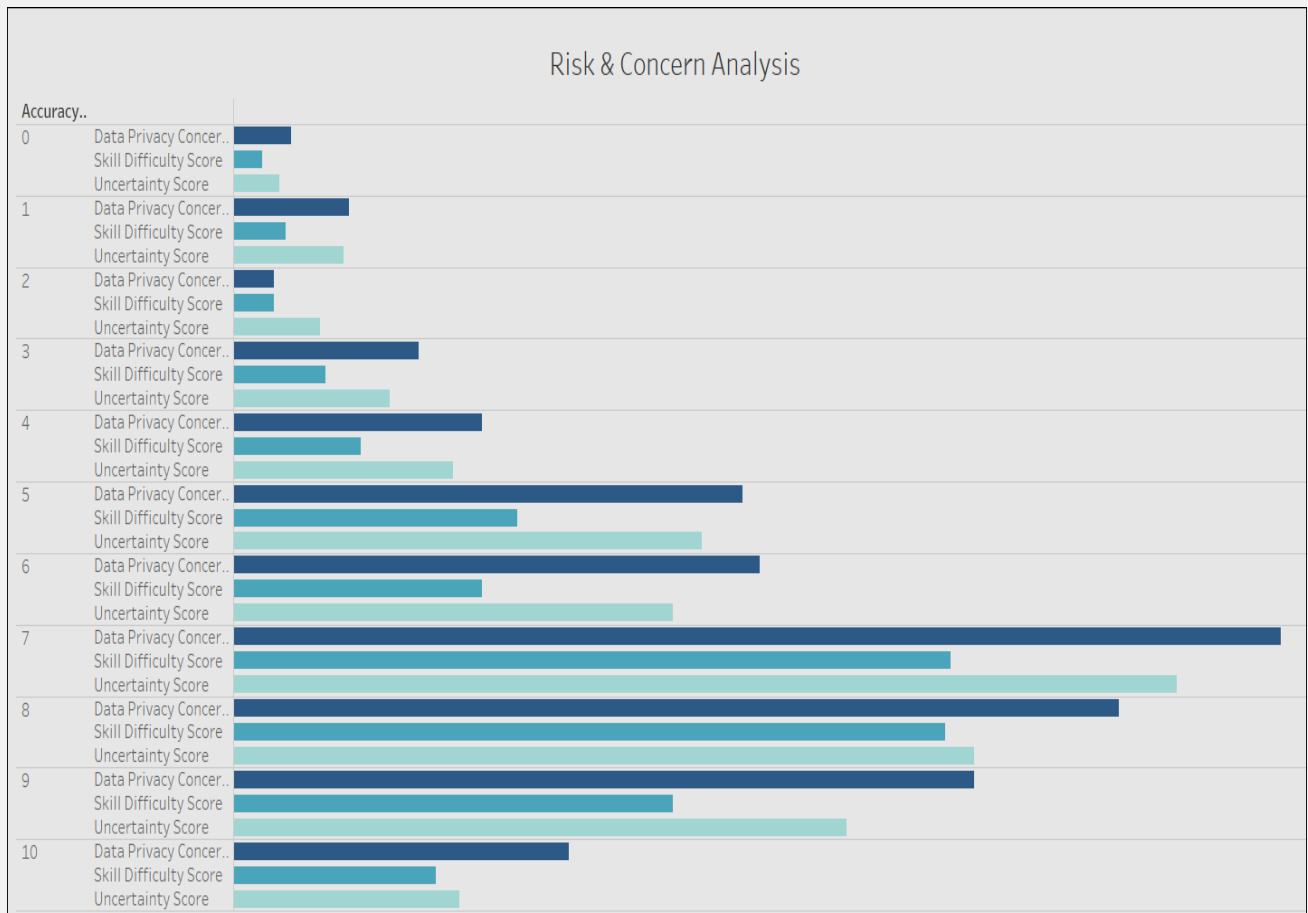
Future enhancements may include collecting a larger and more diverse dataset to improve model generalization. Advanced models such as gradient boosting techniques can be explored to further improve predictive performance. Incorporating real-time data and user feedback, along with additional behavioural and contextual features, could provide deeper insights and strengthen satisfaction prediction accuracy.

Data Analysis and Data Visualization Using Tableau

Tableau Public Link :

https://public.tableau.com/shared/7W5SF7J7Q?:display_count=n&:origin=viz_share_link





The dashboard analysis shows strong adoption and positive impact of generative AI among working professionals in the service industry. A large majority of respondents report being satisfied with AI usage, and most users engage with AI tools on a daily or weekly basis, indicating regular integration into their work routines. The primary use cases include information research, writing support, and technical assistance, highlighting AI's role in knowledge support and task efficiency. Time-saving analysis reveals that most users save up to one hour per day, demonstrating measurable productivity gains. Adoption and perceived benefits are highest among professionals with lower to mid-level work experience, while engagement gradually decreases at higher experience levels. Risk and concern analysis indicates that data privacy, skill difficulty, and uncertainty exist but remain moderate and do not outweigh the overall benefits. Overall, the visual insights confirm that generative AI delivers tangible productivity improvements while maintaining a high level of user satisfaction.

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